

EXPLORATORY DATA ANALYSIS (EDA) AND BUSINESS INTELLIGENCE

Internship Task – 2

Sales & Customer Analytics

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1. Executive Summary

This project performs Exploratory Data Analysis (EDA) and Business Intelligence on a sales dataset. The analysis focuses on data quality, descriptive statistics, customer and product patterns, revenue performance, SQL-based business questions, multivariate relationships, and a KPI dashboard. Excel was used for data preparation, statistical analysis, visualizations, and dashboarding, while MySQL was used for business-oriented data extraction.

2. Objective

To uncover patterns, trends, and relationships within the sales data and develop practical SQL skills for data extraction, aggregation, filtering, and business analysis.

3. Dataset Overview

Item	Details
Records analyzed	1,000
Columns	13
Key numerical fields	Age, Quantity, Unit_Price, Total_Sales
Key categorical fields	Gender, City, Product, Category, Age_Group
Date field	Order_Date

4. Data Preparation & Quality Checks

- Reviewed the dataset for missing values and data completeness.
- Checked for duplicate records and consistency of customer/order information.
- Reviewed data types and formatting for dates, categorical fields, and numerical fields.
- Verified categorical consistency such as Gender and Category values.
- Created a cleaned dataset for subsequent EDA, dashboarding, and SQL analysis.

The cleaned dataset was used as the source for the analysis and MySQL import.

5. Descriptive Statistics & Univariate Analysis

Descriptive statistics were calculated for numerical variables to understand central tendency and spread. Histograms and categorical charts were used to understand distributions.

6. Categorical Analysis

- Gender distribution
- Category distribution
- Product distribution
- City distribution
- Age Group segmentation

7. Revenue & KPI Analysis

KPI	Value
Total Revenue	13,93,99,439
Total Orders	1,000
Total Quantity Sold	5,435

Average Order Value	1,39,399.44
Total Customers	947
Total Cities	8

The dashboard combines these KPIs with Revenue by Category, Revenue by Product, Revenue by City, and Monthly Revenue Trend to provide a compact business view.

8. SQL for Business Questions

Q1. Which products generate the highest total revenue?

```
SELECT Product, SUM(Total_Sales) AS Total_Revenue FROM sales_data GROUP BY Product ORDER BY Total_Revenue DESC;
```

Q2. Which product categories generate the highest revenue?

```
SELECT Category, SUM(Total_Sales) AS Total_Revenue FROM sales_data GROUP BY Category ORDER BY Total_Revenue DESC;
```

Q3. Which cities generate the highest revenue?

```
SELECT City, SUM(Total_Sales) AS Total_Revenue FROM sales_data GROUP BY City ORDER BY Total_Revenue DESC;
```

Q4. What is the monthly revenue trend?

```
SELECT DATE_FORMAT(Order_Date, '%Y-%m') AS Month, SUM(Total_Sales) AS Monthly_Revenue FROM sales_data GROUP BY DATE_FORMAT(Order_Date, '%Y-%m') ORDER BY Month;
```

Q5. How does revenue differ by gender?

```
SELECT Gender, COUNT(DISTINCT Order_ID) AS Total_Orders, SUM(Total_Sales) AS Total_Revenue FROM sales_data GROUP BY Gender;
```

Q6. Who are the top 10 customers by revenue?

```
SELECT Customer_ID, Customer_Name, SUM(Total_Sales) AS Total_Revenue FROM sales_data GROUP BY Customer_ID, Customer_Name ORDER BY Total_Revenue DESC LIMIT 10;
```

Q7. Which products have the highest quantity sold?

```
SELECT Product, SUM(Quantity) AS Total_Quantity_Sold FROM sales_data GROUP BY Product ORDER BY Total_Quantity_Sold DESC;
```

Note: MySQL result tables/screenshots should be inserted below each query in the final submitted version. This report does not invent SQL result values that are not available here.

9. Multivariate Analysis & Correlation

- Correlation analysis among Age, Quantity, Unit_Price, and Total_Sales.
- Scatter plots such as Quantity vs Total_Sales and Unit_Price vs Total_Sales.
- Correlation heatmap to summarize numerical relationships.
- Interpret correlations as associations rather than proof of causation.

10. Business Intelligence Dashboard

A static Excel dashboard was created to summarize key business metrics and revenue patterns. It includes KPI cards and charts for category, product, city, and monthly revenue analysis.

[Insert final Dashboard screenshot here]

11. Key Findings

- The dataset contains 1,000 records across products, categories, cities, customers, and dates.

- Total revenue is approximately ■13.94 crore, with 1,000 total orders and 5,435 units sold.
- The dashboard tracks 947 customers across 8 cities.
- Product, category, and city revenue analysis enables comparison of business performance across segments.
- Monthly revenue analysis helps identify changes in sales performance over time.
- SQL aggregation and sorting provide reproducible answers to business questions.

12. Conclusion

The project demonstrates a complete EDA and basic Business Intelligence workflow: data quality checking, descriptive statistics, categorical analysis, revenue analysis, SQL-based business questions, multivariate analysis, and dashboard creation.

13. Tools Used

- Microsoft Excel – data cleaning, statistics, charts, KPI calculations and dashboard.
- MySQL – data storage and SQL business queries.
- GitHub – project and report submission.
- LinkedIn – final project showcase video.